Gomory cuts

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Solution of the practice quiz

We first write the relaxation of the problem in standard form.

$$\min x_1 - 2x_2$$

subject to

$$\begin{aligned} -4x_1 + 6x_2 + x_3 &= 5, \\ x_1 + x_2 + x_4 &= 5, \\ x_1, x_2, x_3, x_4 &\geq 0. \end{aligned}$$

We solve it using the simplex algorithm, and obtain the following optimal tableau.

x_1	x_2	x_3	x_4		
0	1	0.1	0.4	2.5	x_2
1	0	-0.1	0.6	2.5	x_1
0	0	0.3	0.3	2.5	

The first constraint of the tableau corresponds to a fractional value of x_2. It is written as

$$x_2 + 0.1x_3 + 0.4x_4 = 2.5.$$

The Gomory cut is therefore

$$x_2 + \lfloor 0.1 \rfloor x_3 + 0.4 \lfloor x_4 \rfloor \leq \lfloor 2.5 \rfloor,$$

that is

$$x_2 \leq 2.$$

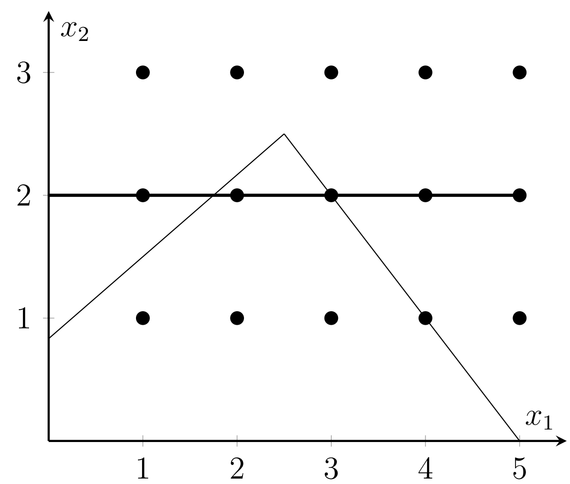

Figure 1: Gomory cut on x_2

This valid inequality is illustrated in Figure 1.

We now introduce the cut on x_2 in the formulation to obtain, after including another slack variable x_5:

$$\min x_1 - 2x_2$$

subject to

$$\begin{aligned} -4x_1 + 6x_2 + x_3 =& 5, \\ x_1 + x_2 + x_4 =& 5, \\ x_2 + x_5 =& 2, \\ x_1, x_2, x_3, x_4, x_5 \geq& 0. \end{aligned}$$

The optimal tableau of this problem is as follows.

x_1	x_2	x_3	x_4	x_5		
0	1	0	0	1	2	x_2
0	0	0.25	1	-2.5	1.25	x_4
1	0	-0.25	0	1.5	1.75	x_1
0	0	0.25	0	0.5	2.25	

Variables x_1 and x_4 are fractional. We consider the constraint associated with x_1 to generate the following cut:

$$x_1 \lfloor -0.25 \rfloor x_3 + \lfloor 1.5 \rfloor x_5 \leq \lfloor 1.75 \rfloor,$$

that is

$$x_1 - x_3 + x_5 \leq 1.$$

In order to draw it, we use the fact that $x_3 = 5 + 4x_1 - 6x_2$ and $x_5 = 2 - x_2$. Therefore, in the original variables, the cut is

$$-3x_1 + 5x_2 \leq 4.$$

This cut is illustrated in Figure 2. It has the property that it cuts the polygon at $(2, 2)$, which becomes a vertex. Actually, the optimal solution of the relaxation of this problem is $(2, 2)$, which is also the optimal solution of the integer optimization problem. No more cuts are necessary.

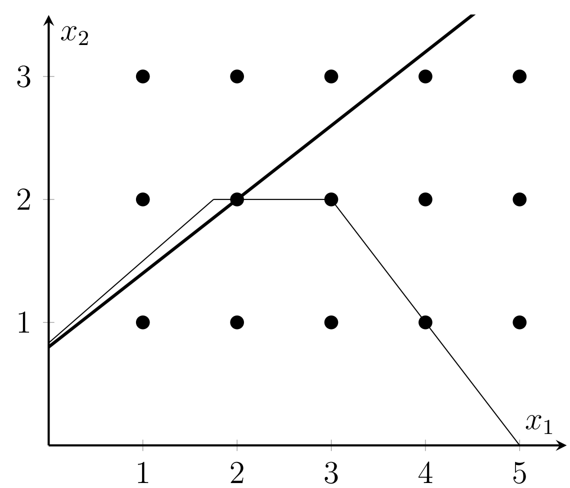

Figure 2: Gomory cut on x_1